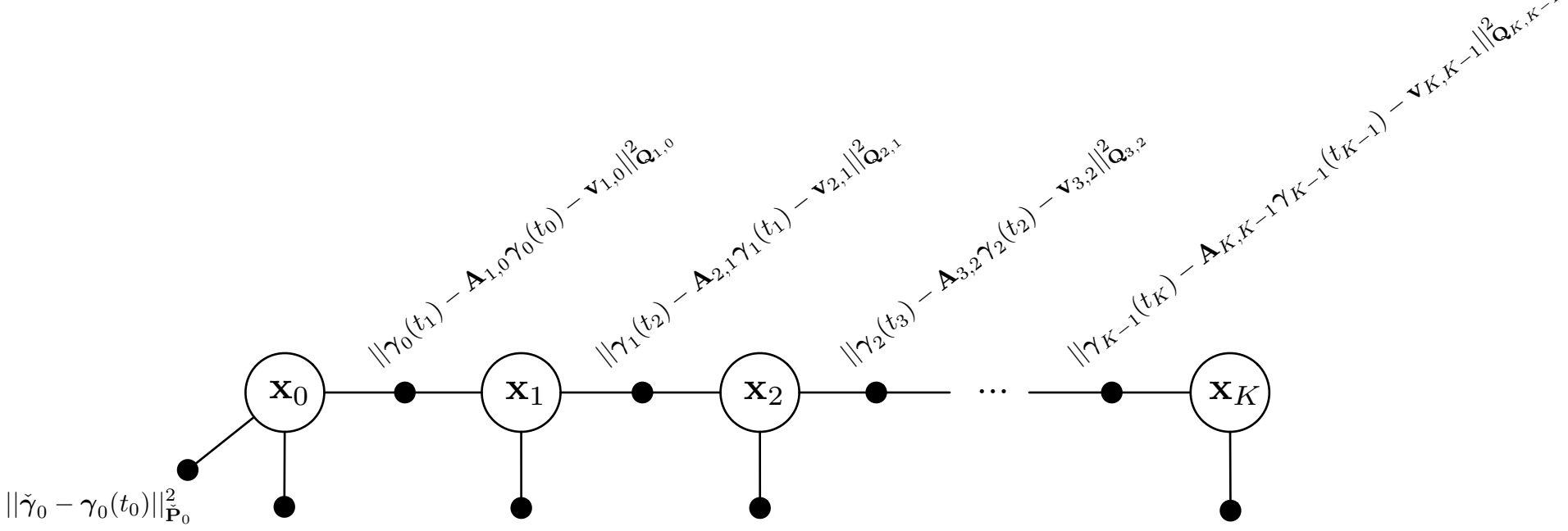

$$\mathbf{x}_k = \{\boldsymbol{T}_k, \boldsymbol{\varpi}_k, \dot{\boldsymbol{\varpi}}_k, \ldots\}$$

$$\boldsymbol{\gamma}_k = \left(\boldsymbol{\xi}_k, \dot{\boldsymbol{\xi}}_k, \ddot{\boldsymbol{\xi}}_k, \ldots\right)$$

$$\boldsymbol{T}(t) = \boldsymbol{T}_k \mathrm{Exp}(\boldsymbol{\xi}_k(t))$$